

Manual

Material Monitoring MPL-MMO 8.2

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1 Material Monitoring

Summary

Purpose

The Material Monitoring function package extends Material and Production Logistics and/or Tracking & Tracing by functions for monitoring minimum storage periods and/or minimum shelf lives and consequently contributes to increasing process security.

Implementation Considerations

You use the Material Monitoring function package if you apply the Material and Production Logistics and/or Tracking & Tracing function packages, and

- need to secure compliance with the shelf lives of incoming raw materials;
- need to secure compliance with minimum shelf lives for semi-finished products and/or WIP material;
- need to secure minimum storage times for semi-finished products and/or WIP material within production.

Integration

This function package is integrated in the function packages Material and Production Logistics as well as Tracking & Tracing, by:

- extending the "batch" object by production and shelf life data;
- verifying the minimum shelf life when incoming batches are registered.

Features

- Monitoring of expiration and warning times
 - Batch and lot-related monitoring of dates and periods
- Warning report
 - Warning report of all lots and batches whose warning date and/or expiration date has been exceeded or will be reached at the evaluation day.
- Expiration statistics
 - Expiration statistics including evaluation as to which material quantities have expired and/or will regularly expire in a given period (table and graph).
- Expiry preview

- Expiry preview including indication as to when which material quantities will have expired, including a forecast of material expiring on basis of the present inventory (table and graph)

2 Applications Available in MOC

Application	Documentation	Transaction code	Function authorization
Warning report	MOC_WarningReport.pdf	warnrp	warnrp
Expiry preview	MOC_ExpiryPreview.pdf	eprev	eprev
Expiry statistics	MOC_ExpiryStatistics.pdf	estat	estat

3 Configuration of Minimum Storage Times and Expiry Durations

Usage

You use minimum storage time to ensure processing-related waiting times within production and to prevent input materials with a minimum storage time that has not expired from being logged on.

You use warning time to be notified in advance in the evaluations [warning report](#) and [expiry preview](#) that material is due to expire.

You use expiry limit to define the maximum storage life of a material.

Requirements

You defined material types for the materials you use.

Procedure

Define on which level the times should be maintained.

- You can update the settings on the material type level in the [material type configuration](#).
- When assigning [material to material type](#), you can overwrite the material type settings for selected materials.

Update the times on the level you have selected.

Results

The times are defined.

4 Activating Material Monitoring

Usage

You activate the material monitoring job if you would like the system to automatically implement the status of known batches in the system based on durations defined in the system.

Requirements

You defined the process time defaults for minimum storage time, warning time and/or expiry time in [material type](#) or in [assignment of material to material type](#).

Procedure

Make an entry in the Scheduler for the cyclic call of the monitoring job and define the call interval, if it is different.

Parameter name	Value
For uploading material receipts:	
Product key	MPL-MMO
License key	MPL-MMO
Command (Windows):	sh.exe ./mpl_lsta.scr
Command (Unix):	./mpl_lsta.scr
Comment:	MPL - Status monitoring
Interval	5



The job is run every five minutes in the system standard delivery. If this interval is sufficient for you, no further adjustments are required.

Results

The system runs the job cyclically at the defined interval.

5 Monitoring Availability Dates

Requirement

You defined the process time defaults for minimum storage time, warning time and/or expiry time in [Material type](#) or in [Assignment of material to material type](#).

You scheduled the [job](#) in the scheduler.

Procedure the system follows during cyclic monitoring

Batches, which because of the *minimum storage time* configuration, were set to status "M" (minimum storage time) at the time the batch was created are set to status "F" (free) once the minimum storage time has expired.

A batch is considered expired if the period of time between when the batch was created and the current time is greater than the period of time defined in the *Expiry limit* configuration. If this is the case, the batch is set to status "V" (expired).

The status modification as a result of the processing step is logged and is shown in the [batch history](#).

Procedure the system follows during input batch logon

When a batch is logged on as an input batch, a plausibility check is run to verify the minimum storage time and the expiry time.

6 Warning Report

Summary

Menu	Material management→ Inventory management→ Warning report
Transaction code	warnrp
Function authorization	Warnrp

Usage

All batches will be shown in the warning report, for which the warning date and/or expiration date has been exceeded or will be reached at the evaluation day. The data display corresponds mostly to the type of display in [batch data maintenance](#).

Selection criteria

The following selection criteria are available in the application:

Batch

Batch number (entry of wildcards possible)

Status

Batch status (selection of several statuses possible)

Material

Article/item (entry of wildcards possible)

Material type

[Material type](#) (entry of wildcards possible)

Material buffer

Material buffer (entry of wildcards possible)

Date, time, relating to

Key date and time related to an expiration date and/or warning time.

Field descriptions

Last modification

- date and time of a batch's last modification
- date of a batch's last modification
- time of a batch's last modification

Modified by

Editor of the last batch's modification

Batch number

Batch number

Quantity

Original quantity

Remaining quantity

Remaining and/or current quantity of the batch

Unit

Quantity unit of the batch

Manufacturing date

Date of creation of a batch

Workplace

Producing workplace of a batch

Batch class

Batch class

Batch information

Information on the batch

Availability date

Date, from which on the batch will be available

Warning date

Warning date of the batch

Warning time

Warning time of the batch

Expiry date

Expiry date of the batch

Status change

Date of the last status change of the batch

Remaining quantity indicator

This indicator shows whether a remaining quantity is assigned to the batch (Y/ N)

Input batch

Input batch number

User

Staff badge number

Batch status

Batch status (free, running, processed, etc.)

Transport status

Transport status of the batch

Workplace

Workplace at which the batch is currently running. Otherwise this is empty.

MES order number

Order number for which the batch is running. Otherwise this is empty.

Reason

Scrap reason

Material

Material number of the batch

Material designation

Designation/ description/ name of the material

Material type

Material type of the batch

Material buffer

Material buffer of the batch

Designation

Designation of the material buffer

Transport unit

Transport unit of the batch

Person

Person, who processed the batch.

Field 1 - 10

Specific batch data (optional)

Production order

Production order, from which the batch was created.

7 Expiry Preview

Summary

Menu	Material management → Inventory management → Expiry overview
Transaction code	eprev
Function authorization	eprev

Usage

The expiry preview deals with this question: "When does how much material expire?" It shows a prognosis of the expired material based on the current stock.

Requirement

All batches that currently have status "F" (Free) or "M" are considered (batch must first be stored for a minimum period before it is available = free). Depending on the selection, consumption is compressed into a specified time frame (daily or hourly).

Selection criteria

The following selection criteria are available in the application:

Material/ material type/ material buffer

Selection of the object to be selected

Material

All materials with the selected material number are selected.

Material type

All data records with the [material type](#) are selected.

Material buffer

All data records in the corresponding [material buffer](#) are selected.

Grid spacing day/ hour

The expiry preview outputs the selection of the grid spacing in days or hours.

Field descriptions

Index

Consecutive number

Date

Date on which the material expires

Quantity

Available batch quantity

Expired quantity

Quantity that expires on the corresponding date

Expired quantity (accumulated)

Total quantity that has expired by this date

Material buffer

Material buffer that includes the expiring material.

8 Expiry Statistics

Summary

Menu	Material management → Inventory management → Expiry statistics
Transaction code	estat
Function authorization	estat

Usage

The expiry statistics displays how much material has expired within a specific period or how much generally expires within a specific period.

Integration

The display of the expiry statistics refers to the material entered, the [material type](#) or [material buffer](#) entered and the period indicated with a grid spacing in the specified cycle. Possible grid values are: 5, 15, 30 minutes, 1, 5, 12 hours, daily, weekly, total.

Selection criteria

The application provides the following selection criteria:

Material

Only the expiry statistics for the selected material number is displayed.

Material type

Only the expiry statistics for the selected [material type](#) is displayed.

Evaluation mode

The evaluation mode indicates the grid spacing of the expiry statistics.

Material buffer

Only the expiry statistics for the selected [material buffer](#) is displayed.

Date from ... to ...

Only the expiry statistics for the selected period is displayed

Field Descriptions

Index

Consecutive number

Evaluation time

Point in time of the material expiration

Material

Material number

Material designation

Material designation

Material type

[Material type](#)

Quantity

Quantity

Unit

Quantity unit

Detail applications**Table display**

In the table display, all of the fields returned by the data source are displayed. A sum is calculated with regard to the quantity. Grouping and subtotals are possible.

Expired quantity per point in time on the grid

Quantity of material that has expired by the respective grid period. The quantity displayed depends on the selection criteria made and can even contain quantities with differing quantity units.

Expired quantity per material in the evaluation period

Quantity of the respective material that has expired in the specified period.